

Food Donation Platform

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Problem statement: Food waste and food insecurity are two critical issues that are present in today's world despite the availability of surplus food from restaurants. Currently, there is no streamlined platform to connect restaurants with surplus food to homeless shelters efficiently. Many shelters struggle to acquire food donations due to logistical challenges, while restaurants are left to discard edible food due to a lack of accessible donation methods. Without a system to assist with real-time donations, food resources are wasted, and food insecurity remains unresolved. Our solution addresses these problems by creating an easy-to-use platform that quickly matches surplus food from restaurants to shelters in need and offers a method for restaurants to donate food immediately.

Value proposition: Our food donation platform bridges the gap between restaurants with surplus food and homeless shelters in need. By automating the donation matching between shelters and restaurants based on matching criteria and location, we ensure compliance with food safety regulations, provide real-time tracking, increase efficiency in food redistribution, reduce waste, and combat food insecurity.

Key features and benefits: We will use AI-driven algorithms to immediately match restaurants with nearby shelters based on food type, quantity, dietary needs, location constraints, and timeframe of necessity, reducing the chance of food waste due to expiration. Our platform will ensure that donations adhere to FDA guidelines and the Good Samaritan Food Donation Act by enforcing restaurants to follow proper policies regarding the handling and distribution of food. Our profile page will provide live updates on donation statuses, ensuring transparency and accountability of both parties. We will have a mobile-responsive web design that allows restaurants and shelters to easily connect and track donations.

Market opportunity: The food waste management industry is valued at about \$52 billion globally. In the U.S., approximately 40% of food goes to waste, while millions experience food insecurity (Gunders, 2012). From this information, it is easy to see that people working in the food waste management industry would be interested in expanding the idea of this application to a broader scale after seeing improvement after use. Our platform addresses the food insecurity gap by offering an easy solution that efficiently redistributes food. Potential users of this application include restaurant owners, shelter coordinators, and food banks, all of whom will benefit from the organization and immediate usability of the application. (Source: Gunders, Dana. "Wasted: How America Is Losing Up to 40 Percent of Its Food from Farm to Fork to Landfill.")

Competitors and competitive advantage:

Competitor #1: Too Good To Go: Too Good To Go is a mobile app that enables restaurants to sell surplus food at discounted prices to consumers. It has a large user base and strong brand recognition. Its focus is on selling rather than donating and it limits the impact it could have on food insecurity. This is not good in cases where the restaurant does not find a buyer and needs to give food out before it expires. Our platform differentiates itself by being donation-driven, directly addressing food insecurity without requiring any method of payment excluding slight transportation costs. (Source: <https://toogoodtogo.com/en>)

Competitor #2: Food Rescue US: Food Rescue US relies on volunteers to transport surplus food from donors to shelters. While it has an established presence and strong partnerships with food donors, its dependency on volunteers for transportation does not allow it to adhere to restaurants that need immediate transportation and are willing to transport. Additionally, shelters with transportation methods can get food quicker if they are allowed to provide their own transportation. Our platform automates donation-matching and tracking, reducing reliance on volunteers and allowing for a faster method of sending food with transportation resources provided by participating users. (Source: <https://foodrescue.us/>)

Key resources: Our platform relies on technology stack that includes React.js for frontend development, Node.js and Express.js for backend operations, Firebase for data storage and authentication, and Google Maps API for location-based matching. The team consists of five computer science majors specializing in software development, UI/UX design, and data analytics. In order to facilitate donation of food, we need to make ourselves known to the FDA and government entities that might run some shelters that will use our application.

Financial summary:

Income: Currently we do not plan on getting income from our users. We may get income in the future from partnerships with non-profits and government agencies, and grants or donations from food security organizations. Since this is a donation platform, the use of any features of the application with a background-checked profile will be free for all users.

Expenses: Expenses will cover web hosting and platform maintenance. Quoting online resources, this will cost approximately \$10 per month or \$120 for a full year. Marketing will cost an average of \$5,000 per year for partnering with organizations like Meta and LinkedIn to suggest the application to businesses. We plan on marketing after gathering donations. Our team is comprised of volunteers and does not require compensation. We will provide the web hosting finances ourselves until we gain donations or grants from supporting organizations.

Financials (USD)	2025	2026	2027	2028	2029
Investments	\$0	\$0	\$0	\$0	\$0
Income	\$0	\$0	\$5,000	\$7,000	\$10,000
Expenses	\$120	\$120	\$5,120	\$5,120	\$7,120
Cash flow (net)	-\$120	-\$120	-\$120	\$1,880	\$2,880

Timeline: In the beginning of 2025 we researched food safety regulations, made a project proposal and user requirement summary, and found improvements to make against competitors. We are currently developing the Figma drafts for our application and plan to finalize them at the end of March. In April 2025, we will develop the application by following the Figma designs and adding the SQL login database and matching algorithm to the platform. In April, we will test the algorithm and begin pilot testing with select restaurants and shelters. We will host our application on a published site beginning May 2025.

Generative AI statement: Generative AI has been used in this assignment.